Exploratory Analysis of Rainfall Data in India for Agriculture using AI & Machine Learning

APSCHE Project Documentation

1. Introduction

Agriculture in India heavily depends on rainfall. Variations in rainfall directly affect crop yield, irrigation planning, and food production. Analyzing historical rainfall data helps farmers and policymakers make better decisions.

This project performs **Exploratory Data Analysis (EDA)** and applies **Machine Learning** techniques to analyze and predict rainfall patterns in India.

2. Objectives

- Analyze historical rainfall data of India
- Identify monthly and yearly rainfall patterns
- Visualize rainfall trends
- Predict annual rainfall using Machine Learning
- Support agricultural planning and water management

3. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook / Google Colab

4. Dataset Description

The dataset contains:

- Year
- Monthly rainfall (January to December)
- Annual rainfall (in mm)

Columns:

YEAR, JAN, FEB, MAR, APR, MAY, JUN,
JUL, AUG, SEP, OCT, NOV, DEC, ANNUAL

5. Data Preprocessing

Steps performed:

- Loaded CSV dataset
- Checked for missing values
- Filled missing values using mean
- Removed duplicate records
- Selected monthly rainfall as input features
- Selected annual rainfall as target variable

6. Exploratory Data Analysis

6.1 Annual Rainfall Trend

A line graph was used to analyze rainfall variation over years.

6.2 Monthly Distribution

Boxplots were used to study rainfall variation for each month.

6.3 Correlation Analysis

A heatmap was used to identify relationships between monthly rainfall values.

7. Machine Learning Model

Model Used:

Linear Regression

Steps:

1. Selected monthly rainfall as features
2. Split data into training and testing sets
3. Trained Linear Regression model
4. Predicted annual rainfall
5. Evaluated model performance

Evaluation Metrics:

- Mean Squared Error (MSE)
- R^2 Score

8. Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
data = pd.read_csv("rainfall_india.csv")

# Data cleaning
data.fillna(data.mean(numeric_only=True), inplace=True)
data.drop_duplicates(inplace=True)

# Features and target
X = data[['JAN','FEB','MAR','APR','MAY','JUN',
        'JUL','AUG','SEP','OCT','NOV','DEC']]
y = data['ANNUAL']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42)

# Model training
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

9. Results

- The model successfully predicts annual rainfall.
- Visualizations helped understand rainfall distribution.
- Correlation analysis showed strong relationships between monsoon months and annual rainfall.

10. Applications in Agriculture

- Crop planning based on rainfall prediction
- Irrigation scheduling
- Drought and flood analysis

- Water resource management

11. Conclusion

This project demonstrates how **Artificial Intelligence and Machine Learning** can be used to analyze rainfall data and support agricultural decision-making. The developed model helps in predicting rainfall and understanding climatic patterns, which is useful for farmers and agricultural planners.

12. Future Scope

- Use Random Forest or Deep Learning models
- State-wise rainfall prediction
- Crop recommendation system
- Drought prediction
- Integration with weather APIs

13. GitHub Structure (Recommended)

```
Rainfall-Analysis/
|— rainfall_india.csv
|— rainfall_analysis.ipynb
|— rainfall_project.pdf
|— README.md
```